

1. Given $x[n] = [1, 2, 3, 4, 4, 3, 2, 1]$
Find radix-2, 8 pt, FFT using DIT & verify your answer using DIF approach.
2. Let $X(k)$ be N -pt DFT of $x(n)$ & N is divisible by 4.
Let $y(n) = x(4n)$; $0 \leq n \leq \frac{N}{4} - 1$ [Decimator]
Express $\frac{N}{4}$ pt DFT of $y(k)$ in terms of $X(k)$.
Hint: Use Modulo/Periodicity property of DFT.
3. Given two complex numbers $(a+jb) \times (c+jd)$. Show that the product of these two complex nos. can be performed with three real multiplications & five additions. Give the algo to support the above observation.
4. As a Dsp engineer you have a number of 8-pt FFT chips available. Show explicitly how you should interconnect three such chips in order to compute a 24-pt DFT.

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